

Shriya G

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 4+ years of combined industry and research experience in developing and maintaining ML pipelines, event-driven microservices using Java, Spring Boot, React, Kafka, and AWS, with a consistent record of optimizing costs and latency.

TECHNICAL SKILLS

- **Programming:** Java, Python, JavaScript, TypeScript, SQL, C#, C, C++
- **Web Technologies & Frameworks:** Spring, Spring Boot, FastAPI, React, Redux, HTML, CSS, JSON, XML
- **Databases:** PostgreSQL, MongoDB, Microsoft SQL Server, MySQL, Oracle, NoSQL
- **Cloud & DevOps:** AWS(EC2, S3, Lambda), Microsoft Azure, Docker, Jenkins, GitHub Actions, CI/CD Pipelines
- **AI/ML:** LangChain, Groq Llama 3.1, XGBoost, SentenceTransformers, RAG Pipelines, Streamlit
- **Software Engineering Practices:** Agile/Scrum, Design Patterns, Responsive Design, Accessibility (WCAG)
- **Tools and Platforms:** Git, GitHub, Figma, Jira, Bitbucket, Jenkins, Postman, Linux

WORK EXPERIENCE

Software Engineer | University of Georgia, Athens, Georgia

Aug 2025 – Present

- Built a GenAI microservice using **LangChain RAG, Groq Llama, and FastAPI** to convert raw ML crop yield metrics into personalized natural-language explanations leveraging Redis caching to optimize response times and reduce LLM API costs
- Architected a hybrid file-storage microservice integrating **AWS S3** via **Strategy Pattern** and **Spring Profiles**, migrating **PostgreSQL BLOBs** to S3 object storage, achieving **70% reduction** in file-storage costs
- Engineered interactive **data visualization dashboard** using **Chart.js** and **Recharts** libraries, implementing 3 chart types (scatter plots, bar and line charts) across 4 **React** components.
- Constructed prediction history tracking system with detailed report generation through **React hooks and API integration**, with downloadable PDF exports.

Graduate Researcher | University of Georgia, Athens, Georgia

Aug 2024 – May 2025

- Developed an end-to-end **machine learning pipeline** to forecast alfalfa biomass yields across four geographic locations using 7K+ weather records over 20 years of historical data, achieving best **R² of 0.907** and **SMAPE of 3.3%**
- Designed **Stacking** and **Voting ensemble** architectures combining 5 **Optuna**-tuned regressors (**XGBoost, Random Forest, Decision Tree, KNN, SVR**), improving prediction accuracy by **59%** over single-model baselines
- Enforced stateless **JWT authentication** with **role-based access control (RBAC)** and **BCrypt** password hashing, ensuring credential security at rest and eliminating server-side session state to support **horizontal scaling** across N service instances.
- Orchestrated end-to-end prediction pipeline with prediction-service and **Kafka**-backed history-service to auto-capture **100%** of ML predictions, eliminating **20% history loss** from failed frontend saves.

Software Engineer I (Full Stack Developer) | *NCR Voyix, Hyderabad, India*

Jan 2022 – Jul 2023

- Developed and maintained 3 multi-tenant **Spring Boot microservices** powering **50+ REST APIs** to **modernize NCR Counterpoint's legacy Delphi POS application**, enabling 50+ external systems to consume POS data over secure JSON endpoints.
- Integrated Spring Boot microservices with **legacy NCR Counterpoint SQL Server** database using **JPA** exposing modern RESTful endpoints while maintaining **100% backward compatibility**
- Established **Kafka-driven asynchronous workflows** to achieve **99.9% system availability** and decrease infrastructure cost by **40%**.
- Achieved **sub-10ms latency (P95)** for API requests by implementing **Redis-based distributed caching** across high-volume NCR Counterpoint microservices, optimizing gift card balance checks and store locator queries.
- Improved test coverage and reliability by implementing **JUnit** and **Mockito** tests, reducing defects by **35%** across core services.
- Collaborated with external integration partners to resolve API issues within **12-hour SLAs**, implementing proactive monitoring and code optimizations that reduced incident volume by **30%**.

Junior Software Engineer | *Pi Software Solutions Private Limited, Hyderabad, India*

Jul 2020 – Dec 2021

- Spearheaded **25+ RESTful APIs** for an **LMS** backend using **Java, Spring Boot, MongoDB, Kafka**, and **Redis**, enabling scalable, tenant isolated exam workflows and analytics.
- Leveraged **Elasticsearch**-powered search, enabling full-text search across 100+ courses with multi-field weighted querying and compound filtering, reducing average search response time by **95%**
- Optimized **Jenkins CI/CD pipeline** reducing test execution time by **40%** through parallel testing across 3 suites and integrated **SonarQube** quality gates with **80% coverage threshold** reducing incidents by **83%**
- **Dockerized** microservices with multi-stage builds, reducing image size by **70%** and enabling **3x faster deployments**
- Documented **70+ REST APIs** using **Swagger/OpenAPI** reducing API integration time **60%** and enabling self-service documentation for frontend developers

PROJECTS

Distributed E-commerce Microservices Platform | *Java, Spring Boot, Kafka, PostgreSQL, Redis* - GitHub

- Built **Spring Boot microservices** with Kafka event-driven workflows, Redis caching, and Spring Cloud Eureka service discovery, enabling asynchronous order processing, autonomous scaling, and fault tolerance across **4+** backend services.

AI-Powered Cold Email Generator | *FastAPI, LangChain, Groq Llama 3.1, ChromaDB RAG* - GitHub

- Constructed **FastAPI** generative AI API with **LangChain**, **Groq Llama 3.1**, and **ChromaDB RAG** that extracts job listings from company career pages, matches relevant portfolio content through semantic search, and generates personalized cold emails with **Pydantic validation** for scalable B2B outreach.

Resume Analyzer | *GenAI RAG Pipeline, Streamlit (UI), LangChain (chains), FAISS (vector DB), Groq Llama 3.1 (LLM), Python*

- Designed scalable **RAG pipeline** with PyPDF2, SentenceTransformers, FAISS indexing, Groq Llama 3.1 analysis via **LangChain**, and **Streamlit UI** and deployed production full-stack app on Hugging Face Spaces.

EDUCATION

University of Georgia, School of Computing, Athens, Georgia

May 2025

Master of Science, Computer Science, CGPA: 3.85/4

CERTIFICATIONS

IBM Full Stack Software Developer Professional Certificate, Coursera – [Credential Link](#)

Microsoft Certified: Azure Fundamentals – [Credential Link](#)

Back-End Development with .NET (Coursera) – [Credential Link](#)

AWS Cloud Practitioner – [Credential Link](#)